

## **STUDIES ON HORMONAL DISORDERS AFFECTING HUMAN FERTILITY**

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### **ABSTRACT:**

The evaluation of the infertile couple is designed to assess the different factors responsible for infertility. For this purpose the study was conducted on about 250 couples at well organized infertility centres based in the city. Out of 250 couples, the disorders were found in 105 females, 85 were males, 35 due to defects in both partners and 25 patients were detected of no cause. This study also included the age range, type and duration of infertility 71.6% of the cases were of primary infertility and 28.4% cases belonged to secondary infertility. The general evaluation of the infertile couple include the detailed history (general, reproductive, sexual, medical) full physical examination, laboratory investigation and diagnostic techniques. The study stressed mainly on the endocrine disorders affecting the fertility. The study conclude that hormones play an important role in the maintenance of fertility of a couple. Therefore the hormonal release in specific magnitude, ratio, sequence and time is very necessary for the adequate function in couples. Any disruption in the delicately coordinated system of hypothalamic- pituitary-ovarian axis may lead to ovulatory dysfunction. The specific diagnosis include no demonstrable cause 9.65%, tubal block 15.2%, Anovulation with regular cycle 7.58%, hyper prolactenemia 13%, PCOD 6.9%, Hirsutism 7.8%, endometriosis 5.5%, and cervical-uterine causes 5.9%.

### **INTRODUCTION**

In order to study the hormonal disorders during human fertility it is necessary to define Infertility, which is explained as failure to conceive after one year of unprotected sexual intercourse. Primary infertility is a situation where no conception has taken place while secondary infertility implies that pregnancy has achieved in the past, even if it may not have yielded a live born body (Mary G. Hummand 1987).

Infertility affects about 10-15% of couples in reproductive age and it is 40% due to defects in male partner, 35% in the female partner, 15% in both partners and 10% due to some unexplained causes (Ladijmi 1987). Hormones play an important role in the fertility of a couple. Their normal synthesis, metabolism and secretion at proper time regulates the cycle and this effects the fertility of a woman. Disturbances in the production of hormones are most frequently clinically apparent in the history and physical examination. The laboratory testing of the endocrine function and the measurement of specific hormones establish the definite diagnosis of infertility.

The evaluation of the infertile couple is designed to assess the factors required for normal fertility. These factors include:

1. Sufficient numbers of progressively motile sperm to penetrate cervical mucus and to reach the ampulla of the fallopian tube to fertilize the oocyte.
2. Normal cervical mucus and uterine and tubal Secretions to support sperm and allow transport.
3. The timely release of a mature oocyte.
4. Uptake and transport of the oocyte by normal fallopian tubes.
5. Normal hormonal production by the ovary to support endometrial development and implantation.

The history, physical examination or both may direct the evaluation in particular directions, such as the assessment of ovulation in women with irregular cycles or laparoscopy, if previous surgery or tubal disease is reported.

The first diagnostic step in the investigation of infertile couple should be the examination of semen. If this is normal then the attention is directed to the woman. Failure to ovulate is the major problem in 40% of woman with infertility, another 30-50% have tubal pathology, and 10% or less have cervical barrier to fertility.

Ovulatory dysfunction is a multifactorial entity. Most methods designed to assess ovulation depend on the detection of progesterone production, which only occurs in high concentrations following the luteinizing hormone surge, ovulation, and corpus luteum formation. The methods include:

1. Basal body temperature
2. Serum progesterone and
3. Endometrial biopsy

Any disruption in the delicately coordinated interaction between the integrated components of the hypothalamic-pituitary-ovarian axis, which must operate within precise quantitative limits and accurate temporal sequences may lead to ovulatory dysfunction. These factors may play an important role in understanding of reproductive processes and future therapeutic modalities.

The hormonal disorders which affect the functioning of ovary or any other which affects fertility are poly cystic ovarian disease (PCOD), Luteal phase defect (LPD), hyper prolactenemia, obesity, endometriosis, Luteinized unruptured follicle syndrome, hirsutism and ovulatory disorders, etc.

## SUBJECT AND METHODS

The research work has been carried out in a well organized infertility centres based in the city. The study group consisted of 250 patients, who sought medical aid for infertility of at least one year. A detailed history (general, reproductive, medical and sexual) was taken from both partners and a full physical examination performed. All the data collected from a particular couple also includes the laboratory investigations and diagnostic techniques.

The study concerned with fertility regulation frequently require estimation of endocrine parameters using Fluoro Immuno Assay (FIA) techniques. Primary Components (antisera, standards, tracers, separating agent) are distributed for the measurement of seven hormones; follicle stimulating hormone (FSH), Luteinizing hormone (LH), prolactin, cortisol, estradiol, progesterone and testosterone. The diagnostic techniques included the hystero salpingography or laparoscopy for tubal patency. A secretory endometrial biopsy performed during the presumptive luteal phase was considered evidence of ovulation. Additional tests were also performed when required.

## RESULTS

By this survey to an infertility centre (Ashfaq Memorial Hospital) we studied 250 couples. Out of 250 couples 105 are females and 85 are males and 35 cases show defects in both, thus 145 are females and 120 are males. Their ages were 18 year to 45 years. These patients had different complaints as: i.e., irregular menstrual cycle (oligomenorrhea or amenorrhea), obesity, weakness, depression stress, itching and aborted or can't maintain pregnancy. A simultaneous diagnosis was reached for the male and female partners in 250 couples (Table 1).

Table 1  
Involvement of each partner in infertility

| Categories                      | No. of Patients | Percentage (%) |
|---------------------------------|-----------------|----------------|
| Female factor only              | 105             | 41             |
| Male factor only                | 85              | 35             |
| Defect in both partners         | 35              | 14             |
| Unexplained (No cause detected) | 25              | 10             |
| Total                           | 250             | 100            |

Thus the male partner is involved in 49% and the female partner in 56% of the cases of infertility.

The data for the age range of the female and male partners, and duration of infertility has been shown in Table No.2. Primary infertility was almost three times (71.6%) of the secondary infertility (28.4%). 80% of couples had a duration of infertility of more than 2 years before coming for investigation work up.

**Table 2**  
**Characteristics of couples and pattern of infertility**

| Characteristics                         | No. | Percentage (%) |
|-----------------------------------------|-----|----------------|
| <b>Age of Wife</b>                      |     |                |
| - < 25                                  | 85  | 34.0           |
| - 26-35                                 | 118 | 47.2           |
| - > 35                                  | 47  | 18.8           |
| <b>Age of Husband</b>                   |     |                |
| - < 25                                  | 28  | 11.2           |
| - 26-35                                 | 139 | 55.6           |
| - > 35                                  | 83  | 33.2           |
| <b>Type of infertility</b>              |     |                |
| - Primary                               | 179 | 71.6           |
| - Secondary                             | 71  | 28.4           |
| <b>Duration of infertility (months)</b> |     |                |
| - 12-24                                 | 57  | 22.8           |
| - 25-48                                 | 83  | 33.2           |
| - > 48                                  | 120 | 48.0           |

The different etiologies of female infertility in our case study show the results in Table 3. The more common cause for infertility at our centre is tubal blockage, which is mostly diagnosed by laparoscopy. This is performed also when no demonstrable cause is present.

**Table 3**  
**Specific diagnosis of infertility (Female Partner)**

| Diagnosis                             | No. | Percentage (%) |
|---------------------------------------|-----|----------------|
| No demonstrable cause for infertility | 14  | 9.65           |
| Tubal block                           | 2   | 15.2           |
| Anovulation with regular cycle        | 11  | 7.58           |
| Anovulation with oligomenorrhea       | 20  | 13.79          |
| Hyper prolactenemia                   | 19  | 13.10          |
| Poly cystic ovarian disease           | 10  | 6.9            |
| Hirsutism                             | 11  | 7.6            |
| Endometriosis                         | 8   | 5.5            |
| LPD with hyper prolactenemia          | 7   | 4.8            |
| LPD with oligomenorrhea               | 9   | 6.2            |
| Cervical, uterine causes              | 8   | 5.5            |
| Ovulatory oligomenorrhea              | 6   | 4.1            |

The details of women who become pregnant and their pregnancy results 163 out of 250 (65.2%) women recruited in the study were successful in conceiving. Out of 170 cases of primary infertility 130 (72.6%) became pregnant whereas 71 cases of secondary infertility 33 (46%) became pregnant.

**Table 4**  
**Pregnancy out come**

| Description                      | No.        |
|----------------------------------|------------|
| Term deliveries                  | 64         |
| Pre term                         | 18         |
| Abortion                         | 14         |
| Ectopic pregnancy                | 3          |
| Continuing pregnancy             | 36         |
| Lost to follow up or transferred | 28         |
| <b>Total</b>                     | <b>163</b> |

Nearly half the patients conceived when the duration of infertility was less than 4 years whereas 22% conceive when the duration is more than 4 years. Almost 7% of

woman become pregnant after only history and examination 60% conceived after initial investigation of endometrial biopsy and some sort of therapy. 10% conceived after the treatment of husband. 40% had term deliveries, 9% was the abortion rate and ectopic pregnancy occurred in 3 cases.

## DISCUSSION

Most studies of infertility have been conducted in clinical settings with persons seeking assistance. Many factors were evaluated for the investigation of the infertile couple. Most common is the ovulatory disorders.

Ovulation is the result of a perfectly balanced and coordinated function of endocrine, paracrine and autocrine systems. Any disruption between the integrated components of the hypothalamic - pituitary ovarian axis may lead to ovulatory dysfunction (Bruno Lunerfeld, 1991). The normal ovarian cycle involves two functions necessary for fertility. The first is the production of mature oocyte, the second is the synthesis of hormones necessary for production of cervical mucus, a secondary endometrium and normal tubal transport (Aral So 1983).

The GnRH, secreted by hypothalamus, in a specific pulsatile manner must act on a properly conditioned pituitary gland, conveyed the message should be correctly received and interpreted by the target cells in the pituitary. This action releases the Gonadotropic hormones, follicle stimulating hormone (FSH) and Luteinizing hormone (LH) in a specific magnitude, ratio, sequence and time. For adequate response LH and FSH must act responsive ovary in a dynamic and predetermined fashion. It has been suggested that a high concentration of LH through the follicular phase allows the developing oocyte to mature prematurely, producing at ovulation an oocyte that is physiologically aged. These oocytes are unlikely to fertilize, or if conception is achieved, early abortion will result (Homberg 1988). Lack of fertilization has been attributed to both over mature and immature oocytes.

Excess of LH or a premature LH surge was believed to lead to follicular luteinization, luteinized unruptured follicle syndrome, or corpus luteum dysfunction (Arumugam et al., 1989). Showed that anovulation with failure of corpus luteum function may be co-existing problem in patients with endometriosis. Endometriosis is a disease affecting fertility in many women. The cause of infertility is obvious when endometriosis results in extensive tubo-ovarian adhesions, tubal distortion, or ovarian endometrioma formation (IVO Brosens, 1991).

Studies have shown that in many cases, other factors and male sub fertility are also involved (Mohmood T.A., 1989).

Many different disease processes can result in a phenotype of hirsutis, anovulation and oligomenorrhea or amenorrhea. Hyper androgenism, the presence of excess androgen production or action, is usually associated with a disordered function of the ovaries, adrenal gland or pilosebaceous unit. Many women with Hyper androgenism receive a diagnosis of "Poly cystic ovary disease (PCOD)" the most common cause of ovulatory disorder and idiopathic hirsutism based on the clinical findings of hirsutism, anovulation, etc. and many are obese (Robert, 1991). Ovulatory failure was found in 89% hirsutism in 69% infertility in 55% to 74% and obesity in 37% to 41% (Raj S.G., 1977). PCOD is characterized by striking abnormalities in levels of LH, FSH, estrogens, androgens, prolactin and glucose metabolism. Significant elevations of virtually all androgens have been found. The principle androgen secreted by the ovary are testosterone and androstenedione.

In general plasma concentration of LH and estrogen are high, FSH are normally low and SHBG is significantly lower. At least 50% patients have elevated DHEA. (SO<sub>4</sub>) and 27% of women have been demonstrated with hyper prolactinemia. Currently, the accepted diagnostic techniques are ultrasonography and laparoscopy. The presence of multiple 2-4 mm cystic structures associated with an increased ovarian stroma is diagnostic (Franks S., 1985).

Failure of ovulation or oligo-ovulation are the main causes of infertility in PCOD. Luteal Phase insufficiency has also been observed in PCOD women having ovulatory cycles (Baird et al., 1981). Hirsutism and acne are clearly the effects of the hyperandrogenemia, which is the characteristic feature of PCOD. Obesity plays a pivotal role in the pathophysiology of PCOD since excess estrone is formed from androstenedione by fat cells (Edman et al., 1978) contributing to the acyclic estrogen milieus. Obesity is related to oligomenorrhea and incidence of oligomenorrhea correlates with the percentage of excess body weight (Hortz et al., 1979). Hyperinsulinemia are also found in PCOD. Persistence of hyperinsulinemia after inhibition of ovarian steroid secretion may modulate ovarian function to overproduce androgens in women with PCOD.

The ovulatory failure in patients with PCOD was due to the mechanical barrier of the thickened capsule. A major reason for ovulatory failure appears to be the low FSH level. The increased LH level will cause stromal hyperplasia in the ovary and combined with accelerated atresia will increase androgen production even more.

Hypersecretion of prolactin is the common cause of pituitary disorder. This condition is roughly found in 20% of women with amenorrhea (peperell, 1981), 15% with anovulatory infertility or may be present as galactorrhea, oligomenorrhea or luteal phase deficiency. Typically patients with hyper prolactinemia have low serum

estradiol levels, low normal LH and FSH, abnormal LH and FSH pulsatility but a normal LH and FSH response to pubatile GnRH therapy (Sander et al., 1984). Infertile women who have elevated PRL levels usually have some evidence of ovulatory dysfunction to explain the infertility. The administration of certain drugs also induce hyper prolactenemia. Hypothyroidism is also associated with increases in serum PRL levels. There are also some physiologic causes of hyper prolactenemia as pregnancy. Ovulation, Stress, Meals, Anesthesia and postictal. The most sensitive and accurate radiologic procedure at this time involves CT scan and MIR (Seang Lin, 1990).

#### ABBREVIATIONS

|      |   |                                   |
|------|---|-----------------------------------|
| FSH  | - | Follicle - Stimulating - Hormone  |
| LH   | - | Luteinizing Hormone               |
| GnRH | - | Gonadotrophin - Releasing Hormone |
| PCOD | - | Poly - Cystic Ovarian Disease     |
| SHBG | - | Sex Hormone Binding Globulin      |
| CT   | - | Computerized Tomography           |
| MIR  | - | Magnetic Image Resonance          |

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